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I build financial software. Shipped, in production, with the tax math unit-tested.

Shipped

NOSTRO — BATCH RECONCILIATION OF FOREIGN REMITTANCES, WITH A MEASURED ACCURACY HARNESS

Solo · TypeScript · Aug–Sep 2026 · Razorpay AI Buildathon, Track 04

Every foreign bank credit an Indian exporter receives must be tied to an invoice *and* a FIRA before the revenue counts as a zero-rated export. Nostro runs that reconciliation as a batch job and reports what it could not resolve as carefully as what it could.

- **250 invoices, twelve noise classes** (split and merged remittances, FX drift across a rate month, flat SWIFT fees, missing certificates, wrong purpose codes). **96.52% allocation precision, 81.62% recall, ~51,000 records/sec.** One command reproduces every number from a seed.
- **₹0.00 under-declared** — the rupee value of anything wrongly called zero-rated. A run that under-declares exits non-zero regardless of its F1. That metric, not accuracy, is the design goal.
- **Ambiguity is refused, never guessed.** 151 exceptions across 7 reason codes, each carrying the exposure it represents, so the queue sorts by what it costs to be wrong about.
- **The layer boundary is enforced by a test:** `src/core/` has no network dependency and cannot import the model layer. The LLM sees only narrations the deterministic parser flags for review, returns a schema-validated hint, and never makes an allocation.
- **110 unit tests**, strict typecheck, none requiring an API key.

RAHA — GST & TAX COMPLIANCE FOR INDIAN CREATORS EARNING FOREIGN INCOME

Solo founder and sole engineer · live at raha.software · Feb–Aug 2026

Indian freelancers and creators who invoice foreign clients must prove each payment is an *export of service* to claim zero-rated GST. Miss the proof and the revenue is reclassified as domestic — 18% GST on money already spent. Raha closes that loop.

- **9,200 lines of TypeScript**, 67 commits, in production on Vercel + Supabase (Mumbai region, for financial data localisation).
- **Every rupee-affecting calculation is a pure, unit-tested function with no I/O** — RBI reference-rate FX conversion, FIRA↔income reconciliation, GST invoice math, advance-tax and GST-liability estimation, export-status classification. Money math does not get to be probabilistic.
- **FIRA reconciliation engine**: matches foreign bank credits to income entries on a $-7/+45$ day window with $\pm 2\%$ amount tolerance (bank charges and forex spreads move the INR figure), handles partial and split allocations, and refuses to assert a match the user has not confirmed.
- **Row-level security on every table**, storage policies for the document vault, Zod validation at every boundary, service-role keys server-side only.
- Next.js 15 (App Router) · TypeScript · Supabase (Postgres + RLS + storage) · Zod · Vitest · Playwright · Tailwind + shadcn/ui.
- Wrote the domain research myself — LUT gating, purpose codes, Sec 2(6) IGST export treatment, the 2026 creator profession code — and sourced it from primary tax guidance, not vibes.

NEXUS — PROVENANCE-AWARE PERSISTENT MEMORY FOR AI SYSTEMS

Solo · research project · Python · 2026

Testing whether a structured, provenance-tracked world model beats conversation history and naive vector retrieval at long-horizon task continuation. Memory records, relationship graph, repositories and lexical retrieval implemented and tested; `mypy --strict` **clean**. Unbuilt subsystems are reserved as empty packages rather than shipped half-done, and the README status table says exactly what is and is not implemented.

How I work with AI

I use AI as an engineering team, not autocomplete — spec, plan, implement, review, ship, with skill libraries and agent harnesses driving the loop. Both projects above were built this way.

The judgment that matters is knowing where **not** to put a model. In Raha, no LLM touches a number that becomes a tax liability. Language models parse messy bank narrations and extract PDF fields; deterministic, tested functions decide what is owed.

Education

Indian Institute of Technology Madras — B.Tech, Chemical Engineering · expected Aug 2028 Relevant coursework: Computational Programming & Process Simulation, Computational Techniques, Statistical Design and Analysis of Experiments, Innovation & Entrepreneurship.

Stack

TypeScript · Python · Next.js · React · Postgres · Supabase · Zod · Vitest · Playwright · Tailwind · Git · Vercel · LLM tooling and agent harnesses